Assessing the Confounding Impact of Socioeconomic Status on the Relationship Between Particulate Matter (PM2.5) and Chronic Respiratory Illness

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Abstract

Respiratory illness doesn't just stem from air quality, but can have determinants from socioeconomic status. Individuals in lower socioeconomic environments are exposed to hazards and pollution rather than people with higher status. This research uses OLS regression to analyze U.S. census tracts by further expanding the surface-level links. Tracking different subdivisions within the U.S. factors infrastructure, industrialization, and air quality. The results of combining the EPA air quality data with the CDC health outcomes and the ACS income metrics changed the project's outlook through an examination of socioeconomic status in relation to environmental determinants. Fine particulate matter and the cause of asthma no longer show statistical significance because of the control of median household incomes. While the air quality in these findings is important, socioeconomic status and population density are also big factors in respiratory illness nationwide.

1. Introduction

Public health research focuses on areas that have high concentrations of PM2.5 or ozone. As a result, air quality can damage a community's health and damage their long-term respiratory health. The environment is the main focus for most epidemiology research, but it is not assessed in neighborhoods. While some projects aim to reduce high concentrations in the air, they don't account for the factors on the ground.

Socioeconomic status factors in the majority of respiratory illnesses because of the different backgrounds faced in different communities. For example, a census tract in a rural city might not have access to healthcare, which may lead to worse health outcomes compared to a smoggy city with more economically stable citizens. This project challenges both aspects of comparison and

allows links between PM2.5 and asthma to be tested when socioeconomic status is taken into account. By connecting system-levels of environmental data and granular income metrics, we can diagnose the main influence of the health inequalities.

2. Methodology

This study used a data channel to conduct a nationwide tract-level analysis by utilizing three distinct public datasets:

- **Health Outcomes (CDC PLACES):** This captures model-based estimates of rates for Current Asthma to understand the current geographic deposition and distribution. Chronic Obstructive Pulmonary Disease (COPD) was extracted by the 2025 release of the CDC PLACES dataset at a census tract level.
- **Environmental Metrics (EPA AQS):** Primarily a repository for ambient air quality data that measures communities and is monitored by state and local agencies. This summarizes annual county-level air quality over an equivalent period. The independent variables were the annual count of days where PM2.5 and Ozone surpassed the baseline limit. These toxins are measured and recorded for the temperature of the air and relative humidity.
- **Socioeconomic Demographics (US Census Bureau):** Provides us with a comprehensive look at the social and economic welfare of a population, which tracks the rates of house ownership, housing value, and cost. The American Community Survey (ACS) 5-Year Estimates API is a survey that provides and collects the median household income data for this project.

2.1 Data Processing and Alignment

The datasets were pivoted to a wider format instead of a long format because isolating asthma and COPD percentages by census tracts allowed the project to change how it stored information. Instead, census tracts were set to their original countries using different standardized FIPS codes and their names. When rows had missing data or reported a negative median income, they were dropped to find the finalized analytical data for the program.

2.2 Model Specification

The severe multicollinearity and condition number instability caused different data and variables to be scaled. In addition, population and income variables were measured differently by a factor of 1,000 to be more accurate. Changing the variable scaling kept the condition number of 1,220 and had highly reliable standard errors, even though there was a big difference in the data between raw population counts and days of poor air quality.

Two Ordinary Least Squares (OLS) regression models were specified:

1. $Asthma_{Pct} \sim Days_{PM25} + Days_{Ozone} + Population\ (Thousands) + Income(Thousands)$
2. $COPD_{Pct} \sim Days_{PM25} + Days_{Ozone} + Population\ (Thousands) + Income(Thousands)$

3. Exploratory Data Analysis

Conducting a bivariate analysis visualized the first relationship between environmental triggers and health outcomes. The raw observations showed that the relationship between PM2.5 and asthma did not provide a positive trendline as expected, but an absent trendline. The data displayed showed a flat relationship that is highly dispersed.

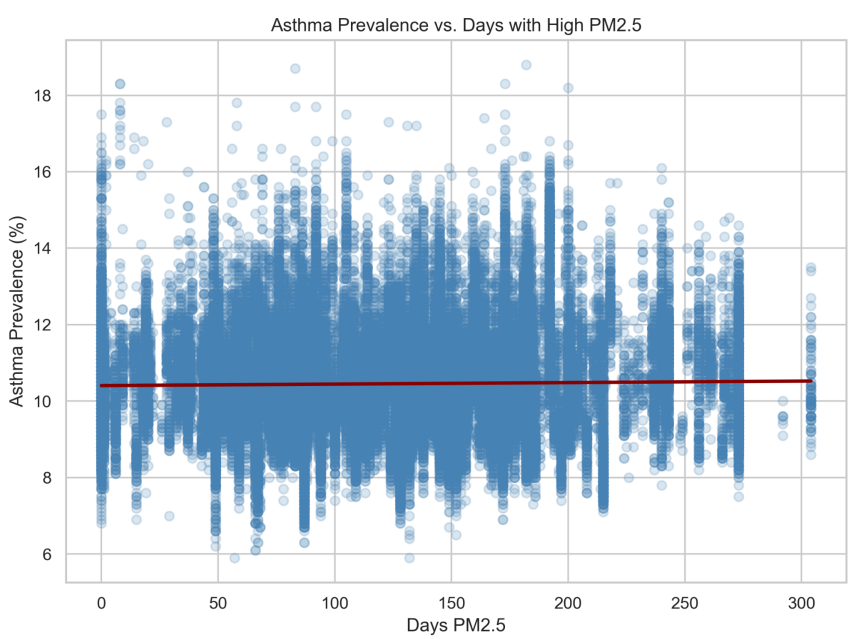

Figure 1: Bivariate scatterplot demonstrating the weak raw correlation between high PM2.5 days and asthma prevalence.

Since the expected positive trendline is absent, and the data represent a highly dispersed, flat relationship, a correlation matrix is added, quantifying all the generating continuous variables.

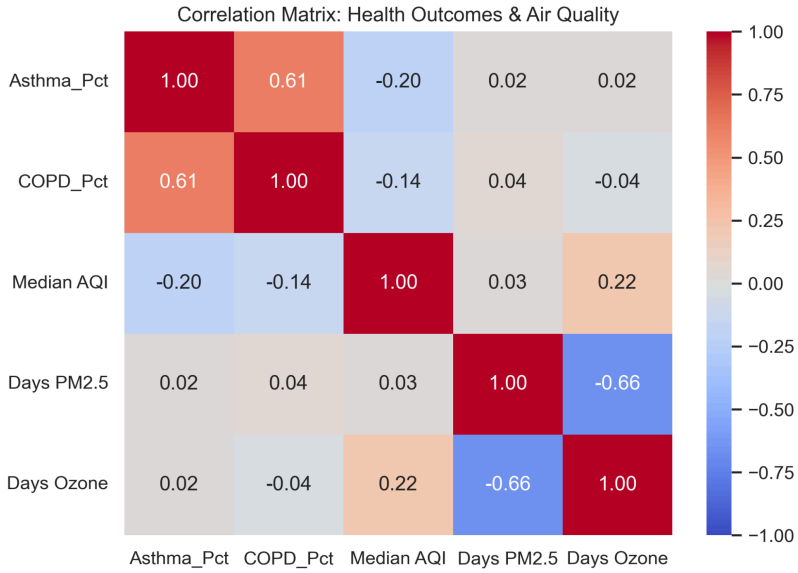

Figure 2: Pearson correlation coefficients across health outcomes and atmospheric metrics.

This correlation matrix reveals the counterintuitive and slightly negative relationship between PM2.5, asthma, and COPD. This suggests that omitting a variable biases the true environmental effect to conclude the socioeconomic controls.

4. Results

To measure the impact of environmental and socioeconomic factors, creating two OLS models helped us better understand and summarize the coefficients and their significance level.

Table 1: Multivariate OLS Regression Results

Variable	Asthma Coeff (p-value)	COPD Coeff (p-value)
Days PM2.5	0.0001 (0.264)	-0.0028 (<0.001)
Days Ozone	0.0019 (<0.001)	0.0011 (<0.001)
Median Income ($k)	-0.0190 (<0.001)	-0.0619 (<0.001)
Population (k)	-0.1017 (<0.001)	-0.2922 (<0.001)
Model R-squared	**0.088**	**0.247**

4.1 Key Findings and Interpretation

The findings in this project reveal and highlight the discrepancy between the two respiratory conditions. The asthma model shows the p-value for PM2.5 (0.267) doesn't contribute to a statistical influence for income being held constant. On the other hand, there was a negative correlation in both conditions with the median household income.

In context, human behavior and their genetics can introduce substantial noise, such as lifestyle and biological variation. This caused the R-squared of 0.247 for the COPD model to become strong. Furthermore, nearly 25 percent of all variation in COPD rates across the nation can be explained by the variables, with the median income being the key variable.

5. Discussion

This research in this environmental epidemiology project doesn't just display how air quality isn't a primary system, but also shows the challenges of the statistical assumptions. While the PM2.5 data fails to predict that asthma is involved does not mean that pollution isn't the main problem. The data concludes that household income has an effect on an individual's respiratory health.

The income of a person's economic status can indicate the environment they live in. For example, someone living in a low-income area can be more exposed to occupational hazards and have poor housing quality, which a county-wide EPA monitor may not take into account. Some monitors may not pick up data from a massive highway that is considered a major trigger. However, wealthy neighborhoods in a smog county may still have results lower than those of a cleaner area in respiratory health. In addition, the data presented in this research informs us that environmental regulations might be looking past certain determinants if they don't account for socioeconomic status in specific census tracts.

6. Conclusion

The analysis shows that having multivariate, cross-domain datasets in public health analytics is important for conducting research. The data provides a view of environmental risks in a county nationwide and reveals that chronic respiratory illnesses like asthma and COPD are prevalent. The regression models presented confirm that the reliability of median household income as a predictor for respiratory health is higher than isolated county-level air quality metrics. The models in future epidemiological modeling must prioritize the other factors in environmental research, like socioeconomic status, to allocate medical resources accurately.